

Shuchang Xu

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Education

The Hong Kong University of Science and Technology	Hong Kong, China
Ph.D. in Computer Science, Advisor: Dr. Huamin Qu	Aug 2023 - Present
Tsinghua University	Beijing, China
M.S. in Computer Science, Advisor: Dr. Yuanchun Shi	Aug 2018 - Jun 2021
Tsinghua University	Beijing, China
B.S. in Electrical Engineering, GPA: 91.2/100 (Top 5%)	Aug 2014 - Jun 2018
B.S. in Digital Media Art and Design (Second Major)	

Research Overview

As a systems researcher in Human-Computer Interaction, I create **AI-powered interactive systems that enhance visual accessibility and support human flourishing** in three core areas:

- **Multi-Modal AI for Accessible Visual Media:** I build multi-modal AI pipelines that enable blind users to engage with diverse visual media, including 360° videos [1], 2D videos [2], and photo collections [3].
- **Multi-Sensory Interfaces for Safe Navigation:** I create wearable interfaces that enhance safe navigation for blind users through on-body haptics [4], head-mounted light cues [5], and handheld vibrations [6].
- **Human-AI Systems for Learning Support:** I co-design human-AI systems that support learning by facilitating problem-solving [7] and skill acquisition [8].

Conference Publications

- [1] **Branch Explorer: Leveraging Branching Narratives to Support Interactive 360° Video Viewing for Blind and Low Vision Users.**
Shuchang Xu, Xiaofu Jin, Wenshuo Zhang, Huamin Qu, Yukang Yan.
UIST 2025 - ACM Symposium on User Interface Software and Technology.
- [2] **DanmuA11y: Making Time-Synced On-Screen Video Comments (Danmu) Accessible to Blind and Low Vision Users via Multi-Viewer Audio Discussions.**
Shuchang Xu, Xiaofu Jin, Huamin Qu, Yukang Yan.
CHI 2025 - ACM Conference on Human Factors in Computing Systems.
- [3] **Memory Reviver: Supporting Photo-Collection Reminiscence for People with Visual Impairment via a Proactive Chatbot.**
Shuchang Xu, Chang Chen, Zichen Liu, Xiaofu Jin, Linping Yuan, Yukang Yan, Huamin Qu.
UIST 2024 - ACM Symposium on User Interface Software and Technology.
- [4] **Virtual Paving: Rendering a Smooth Path for People with Visual Impairment through Vibrotactile and Audio Feedback.**
Shuchang Xu, Ciyuan Yang, Wenhao Ge, Chun Yu, Yuanchun Shi.
IMWUT 2020 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.
- [5] **LightGuide: Directing Visually Impaired People along a Path Using Light Cues.**
Ciyuan Yang, Shuchang Xu (co-first), Tianyu Yu, Guanhong Liu, Chun Yu, Yuanchun Shi.
IMWUT 2021 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.

- [6] **Tactile Compass: Enabling Visually Impaired People to Follow a Path with Continuous Directional Feedback.**
Guanhong Liu, Tianyu Yu, Chun Yu, Haiqing Xu, Shuchang Xu, Ciyuan Yang, Feng Wang, Haipeng Mi, Yuanchun Shi.
CHI 2021 - ACM Conference on Human Factors in Computing Systems.
- [7] **NeuroSync: Intent-Aware Code-Based Problem Solving via Direct LLM Understanding Modification.**
Wenshuo Zhang, Leixian Shen, Shuchang Xu, Jindu Wang, Jian Zhao, Huamin Qu, Linping Yuan.
UIST 2025 - ACM Symposium on User Interface Software and Technology.
- [8] **RhythmTA: a Visual-Aided Interactive System for ESL Rhythm Training via Dubbing Practice.**
Chang Chen, Sicheng Song, Shuchang Xu, Zhicheng Li, Huamin Qu, Yanna Lin.
UIST 2025 - ACM Symposium on User Interface Software and Technology.
- [9] **Accurate and Low-Latency Sensing of Touch Contact on Any Surface with Finger-Worn IMU Sensor.**
Yizheng Gu, Chun Yu, Zhipeng Li, Weiqi Li, Shuchang Xu, Xiaoying Wei, Yuanchun Shi.
UIST 2019 - ACM Symposium on User Interface Software and Technology.
- [10] **A Machine Learning Framework to Identify Routing Short Violations from a Placed Netlist.**
Aysa Tabrizi, Nima Darav, Shuchang Xu, Logan Rakai, Ismail Bustany, Andrew Kennings, Laleh Behjat.
DAC 2018 - ACM/IEEE Design Automation Conference.

Poster/Demo Publications

- [a] **MemeSpeak: Making Image Memes Blind-Accessible through Emotional Speech Augmentation.**
Gengyu Chen, Chiqian Xu, Junru Jiang, Yuting Lu, Xiao Wang, Huamin Qu, Shuchang Xu, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.
- [b] **From Audio Description to Movement: Challenges and Design Principles in Video-based Body Exercise for Blind and Low Vision Users.**
Ziyi Jiang, Mengjie Jian, Jiajia Hu, Hongze Zhao, Huamin Qu, Shuchang Xu, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.
- [c] **When Accessibility Becomes a Trap: A User-Centric Characterization of Dark Patterns Arising from Screen Reader Users' Perceived Deception in Mobile Interfaces.**
Dai Shi, Lin Tian, Jiaxun Sun, Tomimi Kawakami, Nuo Cheng, Shuchang Xu, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.

Fellowship & Awards - Ph.D.

Hong Kong Postgraduate Fellowship	2024–2027
Full stipend, approximately USD 50,000 per year	
Best Paper Honorable Mention Award	Apr 2025
CHI 2025, among the top 5% out of 5020 submissions	
HKUST RedBird Academic Excellence Award	2024–2025
HKD 40,000 for outstanding academic and research performance	
Research Travel Awards	2024–2025
HKD 14,000 for CHI 2025; HKD 14,700 for UIST 2024	
Special Recognition for Outstanding Reviews	
CHI 2025, IMWUT 2025, ISMAR 2025, CHI 2025 LBW	

Fellowship & Awards - Master & Undergraduate

Outstanding Graduate in Computer Science, Tsinghua University	Jun 2021
China National Scholarship for Postgraduates (Top 1%)	Nov 2020
China National Scholarship for Postgraduates (Top 1%)	Nov 2019
China National Scholarship for Undergraduates (Top 1%)	Nov 2017
Xiaomi Artificial Intelligence Hackathon, 2nd Prize	Jul 2022
Best Paper Award for HCI in Games, HCI International 2019	Jul 2019
Microsoft Mixed Reality Development Contest, 2nd Prize	Dec 2017

Teaching Experiences

Guest Lecturer , User Experience Design, Tongji University	2025
Teaching Assistant , Java Programming Language, HKUST	2025
Teaching Assistant , User Experience Design, Tsinghua University	2020
Teaching Assistant , Big Data and Machine Intelligence, Tsinghua University	2019

Invited Talks

UIUC SALT Lab , Invited by Dr. Yun Huang	Jun 2025
Improving Visual Media Accessibility with AI-Assisted Interactive Storytelling Techniques	
Tongji University , Invited by Dr. Guanhong Liu	Apr 2025
AI-Powered Interactive Systems for Improving Visual Media Accessibility	

Professional Experiences

Xiaomi – Software Product Manager – Beijing, China	Aug 2021 - Jun 2023
Bytedance – TikTok Technical Artist (Intern) – Shanghai, China	Jun 2020 - Aug 2020
Tencent – Game Level Designer (Intern) – Shenzhen, China	Jun 2019 - Aug 2019
NetEase – Game Interaction Designer (Intern) – Hangzhou, China	Jun 2018 - Aug 2018

Patents

Vision Accessibility Techniques: CN111494175
Mobile Game Interaction Techniques: CN109316745, CN109460179, CN109550242
Smartphone Interaction Techniques: CN115857375, CN115755635

Skills

Programming: C#, Swift, Java, Python, React Native, C++, OpenGL, OpenCV
Platforms/IDE: Unity3D, Xcode, Android Studio
Prototyping: Hardware (Arduino, Raspberry PI); 3D (3ds Max, Maya), UX Design (After Effects, Figma)
TOEFL: 109/120 (R 29/30, L 26/30, S 24/30, W 30/30)